

Occlusion-Robust LiDAR Human Segmentation via Geometry-Consistent Occlusion Synthesis

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Abstract—Human segmentation in LiDAR depth images is challenging due to occlusions and incomplete measurements in indoor environments. In this paper, we propose a LiDAR-based framework for occlusion-robust human segmentation that reconstructs plausible complete human masks from heavily occluded depth images. Our approach introduces a geometry-consistent LiDAR occlusion synthesis pipeline for data augmentation, enabling realistic synthesis of occluded depth data with automatic annotations. In addition, we propose a human-prior-guided segmentation model that integrates 3D pose priors and modal masks extracted from occluded depth data within a U-Net framework. Experiments show significant improvements over a strong U-Net baseline, increasing the Dice score from 64.55% to 92.35% and reducing the L1 error from 14.52% to 1.69% under severe occlusions. Qualitative results on real-world data further confirm the effectiveness of the proposed approach.

Index Terms—LiDAR, human segmentation, occlusion synthesis, human sensing, deep learning

I. INTRODUCTION

LiDAR sensors have recently attracted increasing attention for indoor Human Activity Recognition (HAR) thanks to their privacy-preserving nature and their ability to provide reliable depth information. However, depth measurements acquired by LiDAR sensors are often affected by sensor noise and partial occlusions caused by surrounding objects, which may severely degrade the performance of HAR systems. Occlusion occurs when an object partially blocks the human body, leading to missing or corrupted information. Therefore, recovering plausible complete human masks from occluded LiDAR depth images can improve the robustness of HAR systems in realistic indoor environments where occlusions frequently occur [1].

Human segmentation under occlusion has been extensively studied in RGB images. Several works have shown that pose-guided methods and approaches relying on modal masks, which represent only the visible regions of the human body while excluding occluded parts, are effective for recovering missing body parts under occlusion [2], [3].

In contrast, existing works on LiDAR-based depth imaging have mainly addressed 3D pose estimation for HAR, often under simplified conditions such as top-view sensor setups that inherently avoid occlusions [1], or assuming largely visible subjects [4]. More recently, transfer learning has been investigated for automatic label generation in LiDAR-based

pose estimation and segmentation tasks [4], [5]. However, these approaches do not explicitly address complete human segmentation under severe occlusions.

Regarding segmentation directly, existing methods for LiDAR depth images remain limited to fully visible humans and rely on standard architectures such as U-Net [4]. Moreover, although synthetic occlusion techniques are widely used in RGB domains [2], [6], realistic occlusion modeling for LiDAR depth images with geometry-consistent object insertion has received limited attention. Since obtaining large-scale annotated LiDAR data under occlusion is particularly difficult, realistic occlusion synthesis becomes crucial for generating representative training data.

Therefore, to address these limitations, we propose a framework that combines a LiDAR-specific geometry-consistent occlusion synthesis pipeline with a human-prior-guided segmentation model. The main contributions of this work are summarized as follows:

- 1) **LiDAR-based human segmentation under occlusion:** we present a novel approach for recovering plausible complete human masks from severely occluded LiDAR depth images.
- 2) **Geometry-consistent LiDAR occlusion synthesis:** we propose a novel data augmentation pipeline specifically designed for LiDAR depth sensors. Our approach uses a 3D CAD (Computer-Aided Design) object insertion strategy to synthesize realistic, depth-consistent occlusions and generate large-scale annotated training data.
- 3) **Human-prior-guided segmentation model:** we introduce a segmentation model that jointly incorporates human pose and modal mask priors to improve human segmentation under severe occlusions. Experimental results show that these priors provide complementary gains on top of the proposed geometry-consistent occlusion synthesis pipeline.

II. METHOD

We first describe the data acquisition and automatic annotation process from aligned RGB and LiDAR data. We then present our geometry-consistent LiDAR occlusion synthesis pipeline for data augmentation. Finally, we present the proposed human-prior-guided segmentation model.

A. Data Acquisition and Automatic Ground-Truth Generation

To construct the dataset, we use the Intel RealSense LiDAR L515, which provides synchronized RGB and depth streams at 1024×768 resolution and 30 fps. Depth maps are converted into metric distances and aligned with RGB images using the RealSense SDK, ensuring pixel-wise correspondence. Ground-truth human masks are automatically generated from RGB images using Mask2Former [7], following the pipeline in [4]. Thanks to the pixel-wise RGB-depth alignment, these masks are directly transferred to the depth maps without manual annotation. Similarly, following the strategy proposed in [5], 3D pose annotations are automatically generated using a top-down pipeline composed of RT-DETR [8] and ViTPose [9]. 2D keypoints are estimated on RGB images and lifted to 3D using aligned depth values. Therefore, pixel-wise RGB-depth alignment, combined with RGB pretrained models, enables the automatic generation of accurate mask and pose annotations without manual labeling.

B. Geometry-Consistent LiDAR Occlusion Synthesis

To address the lack of annotated occluded LiDAR depth data, we propose a geometry-consistent LiDAR occlusion synthesis strategy using 3D CAD objects from the ModelNet40 dataset [10]. Unlike conventional random erasing or copy-paste augmentation [2], [6], our method preserves the depth ordering and geometric consistency of LiDAR measurements, producing physically plausible occlusions.

The proposed synthesis pipeline proceeds as follows. First, a 3D CAD object is randomly sampled and transformed using random scaling and rotation to generate diverse occluding objects. The transformed mesh is then projected into the scene according to the intrinsic parameters of the Intel RealSense L515 camera [11]. To generate realistic occlusions, a human pixel is randomly selected from the input LiDAR depth image, and its depth value is used to place the object slightly closer to the sensor. The object position is constrained to avoid unrealistic intersections with the human body, resulting in meaningful partial occlusions. In a small fraction of samples, the object is instead placed on background regions to increase scene diversity. After rendering the 3D object into a depth map, the occlusion ratio is computed, and the object placement is repeated until the desired occlusion level is reached. Finally, the rendered depth map is merged with the original scene using z-buffering, which preserves the closest surface at each pixel. This procedure produces physically plausible occluded LiDAR depth images.

C. Human-Prior-Guided Segmentation Model

Our second contribution is a human-prior-guided segmentation model designed for LiDAR depth images. Inspired by [2], [3], we incorporate two complementary priors: a 3D pose prior and a modal mask prior, which captures only the visible human region under occlusion. The proposed pipeline consists of two stages.

In the first stage, we estimate 3D human keypoints from depth images using an adapted ViTPose backbone. We build

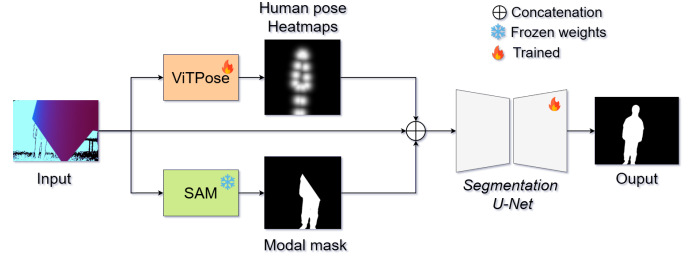


Fig. 1: Overview of the proposed human-prior-guided segmentation model. The occluded LiDAR depth image, estimated 3D pose heatmaps, and modal mask prior are concatenated and fed into a U-Net to recover the complete human mask.

upon a pretrained RGB ViTPose model and transfer its knowledge to LiDAR data using Low-Rank Adaptation (LoRA) [12], which is inserted into the self-attention layers. During training, only the LoRA parameters and a lightweight prediction head are updated, preserving the strong pose priors learned from large-scale RGB datasets while enabling efficient adaptation to LiDAR inputs. We normalize depth maps and convert them into pseudo-RGB representations before feeding them into the ViTPose network. The modified head predicts 34 heatmaps: 17 heatmaps for 2D joint localization and 17 additional heatmaps encoding the corresponding depth (z -coordinate), resulting in a full 3D pose representation. We also use the pre-trained Segment Anything Model (SAM) [13] in a zero-shot manner, prompted by the bounding boxes from RT-DETR [8], to automatically extract modal masks from the same pseudo-RGB depth images. Although SAM is trained on RGB images, we empirically observed that it produces sufficiently accurate modal masks on LiDAR depth images. These masks serve as a reliable modal prior, providing valuable structural cues about the visible human shape under occlusion.

In the second stage, the estimated 3D pose heatmaps and modal mask are concatenated with the occluded LiDAR depth image and fed into a U-Net-based segmentation network to recover the complete human mask. Finally, a connected component analysis is applied as a post-processing step to refine the predicted mask by removing isolated regions. An overview of the proposed human-prior-guided segmentation model is illustrated in Fig. 1.

III. EXPERIMENTAL EVALUATION

A. Dataset and Implementation details

We collected 14,040 frames from 7 subjects (4 males and 3 females) performing diverse indoor activities. To avoid data leakage, the original frames are first divided into training, validation, and test sets at the subject level. The proposed occlusion synthesis augmentation is then applied independently within each split. By generating multiple augmented samples per frame, the final dataset contains 38,950 training, 1,106 validation, and 4,336 test samples. The test set is further divided into weak (0–20%), moderate (20–40%), and severe (40–60%) occlusion levels, and includes unseen subjects and

TABLE I: Ablation study under weak (W), moderate (M), and severe (S) occlusions. Synt. denotes occlusion synthesis.

Method	Occl.	Dice $\uparrow$	IoU $\uparrow$	inv-IoU $\uparrow$	L1 $\downarrow$
Depth (Baseline U-Net)	W	81.18	71.20	31.06	5.55
	M	74.55	63.25	48.38	9.30
	S	64.55	51.53	43.58	14.52
Depth + Synt.	W	96.03	92.58	88.28	0.87
	M	94.42	89.76	85.25	1.26
	S	91.58	85.05	81.16	1.90
Depth + Synt. + Modal	W	96.16	92.76	88.07	0.83
	M	94.53	89.91	85.34	1.21
	S	91.71	85.17	81.27	1.84
Depth + Synt. + Pose	W	95.75	92.19	87.90	0.95
	M	94.14	89.44	85.13	1.33
	S	91.67	85.44	82.44	1.91
Depth + Synt. + Modal + Pose	W	96.22	92.87	87.93	0.79
	M	94.62	90.06	85.37	1.15
	S	92.35	86.21	82.89	1.69

occluding objects from ModelNet40. A separate real-world dataset of 321 frames from an unseen subject was collected under real occlusions. Since dense ground-truth annotations are unavailable for real occluded scenes, the real-world dataset is used only for qualitative evaluation.

The segmentation network is implemented in PyTorch and trained with the Adam optimizer [14] using a learning rate of 10^{-4} , weight decay of 10^{-8} , batch size of 32, and a combination of Binary Cross-Entropy (BCE) and Dice loss. The ViTPose-LoRA model is trained with AdamW [15] and Mean Squared Error (MSE) loss using a learning rate of 10^{-3} and weight decay of 10^{-4} . LoRA with rank $r = 16$ is applied to the self-attention layers. All data is resized to 512×288 . Experiments were conducted on a single NVIDIA RTX 3090 GPU.

The models are evaluated using Dice, IoU, inv-IoU (IoU computed on occluded regions), and L1 error.

B. Ablation Study

Table I presents an ablation study under different occlusion levels. First, introducing the proposed geometry-consistent occlusion synthesis (+Synt.) substantially improves the performance compared with the standard baseline (U-Net [4]), increasing the Dice score from 64.55% to 91.58% under severe occlusions while reducing the L1 error from 14.52% to 1.90%. These results indicate that realistic occlusion simulation is the primary factor contributing to segmentation robustness under severe occlusions. Second, both priors provide complementary gains over Depth + Synt. The modal mask prior better preserves shape information under weak and moderate occlusions, while the pose prior is more effective at recovering heavily occluded body parts, as reflected by its higher inv-IoU under severe occlusions. Finally, combining both priors yields the best overall performance. The largest gains are observed under severe occlusions, where our full model achieves a Dice score of 92.35%, an IoU of 86.21%, and an inv-IoU of

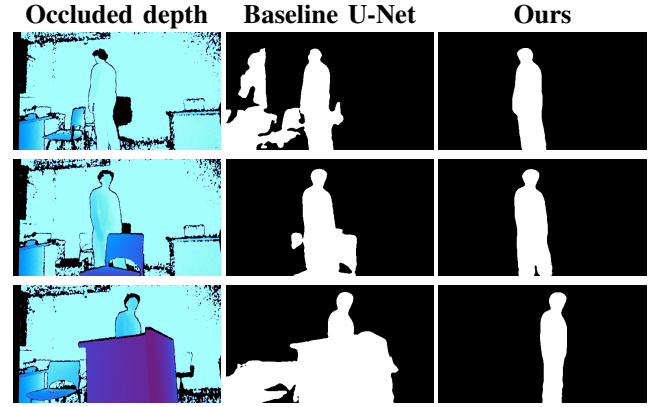


Fig. 2: Qualitative comparison between the standard baseline (U-Net [4]) and our proposed method (Ours) on real occluded LiDAR depth images.

82.89%, while reducing the L1 error to 1.69%. These results demonstrate the complementary benefits of pose and modal mask priors for recovering heavily occluded human shapes.

C. Visual Results

Fig. 2 shows a comparison between the standard baseline (U-Net [4]) and our proposed framework on real-world occluded LiDAR depth images under weak, moderate, and severe occlusion levels. In all cases, the baseline U-Net is easily affected by non-human artifacts, especially under severe occlusions. In contrast, our proposed method visually recovers more plausible complete human masks, suggesting its robustness on real-world data. This improvement can be attributed to the proposed occlusion synthesis pipeline combined with strong human-centric priors, which help the model focus on relevant human regions and reduce confusion from background structures.

IV. CONCLUSION

In this paper, we proposed a LiDAR-based framework for occlusion-robust human segmentation that combines geometry-consistent occlusion synthesis with a human-prior-guided U-Net. Experimental results demonstrate substantial improvements over a strong U-Net baseline. Under severe occlusions, the proposed framework improves the Dice score from 64.55% to 92.35% while reducing the L1 error from 14.52% to 1.69%, demonstrating the effectiveness of combining realistic LiDAR occlusion synthesis with human priors.

Although the proposed framework generalizes well to real-world indoor LiDAR scenes and runs at 8.6 FPS, its applicability to outdoor environments and different LiDAR sensors remains to be explored. Future work will investigate model optimization for real-time HAR deployment and LiDAR-specific noise modeling to improve occlusion realism.

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REFERENCES

- [1] X. Wu, H. Zhang, C. Kong, Y. Wang, Y. Ju, and C. Zhao, "Lidar-based 3-d human pose estimation and action recognition for medical scenes," *IEEE Sensors Journal*, vol. 24, no. 9, pp. 15 531–15 539, 2024.
- [2] A. Dutta, R. Lal, D. S. Raychaudhuri, C.-K. Ta, and A. K. Roy-Chowdhury, "Poise: Pose guided human silhouette extraction under occlusions," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Waikoloa, HI, USA, 2024, pp. 6141–6151.
- [3] Q. Zhou, S. Wang, Y. Wang *et al.*, "Human de-occlusion: Invisible perception and recovery for humans," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 3691–3701.
- [4] M. Chebib, M. Bouazizi, and T. Otsuki, "Segmentation-guided denoising and gradient-aware depth completion for enhanced lidar-based human activity recognition," in *Proceedings of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2025.
- [5] X. Meng, M. Bouazizi, Z. Li *et al.*, "Indoor human activity recognition using multiple dynamic nonlinear mapping applied to 3-d lidar-collected data," *IEEE Internet of Things Journal*, vol. 12, no. 11, pp. 16 797–16 812, 2025.
- [6] T.-Y. Lin, M. Maire, S. Belongie *et al.*, "Microsoft coco: Common objects in context," in *European Conference on Computer Vision (ECCV)*, 2014.
- [7] B. Cheng *et al.*, "Mask2former for video instance segmentation," *arXiv preprint arXiv:2112.10764*, 2021.
- [8] Y. Zhao, W. Lv, S. Xu *et al.*, "Detrs beat yolos on real-time object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [9] Y. Xu, J. Zhang, Q. Zhang *et al.*, "Vitpose: Simple vision transformer baselines for human pose estimation," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [10] Z. Wu, S. Song, A. Khosla, F. Yu, L. Zhang, X. Tang, and J. Xiao, "3d shapenets: A deep representation for volumetric shapes," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2015.
- [11] A. Breitharth, C. Hake, and G. Notni, "Measurement accuracy and practical assessment of the lidar camera intel realsense l515," in *Optical Measurement Systems for Industrial Inspection XII, SPIE*, 2021.
- [12] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," *arXiv preprint arXiv:2106.09685*, 2021.
- [13] A. Kirillov, E. Mintun, N. Ravi *et al.*, "Segment anything," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [14] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.
- [15] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," *arXiv preprint arXiv:1711.05123*, 2017.